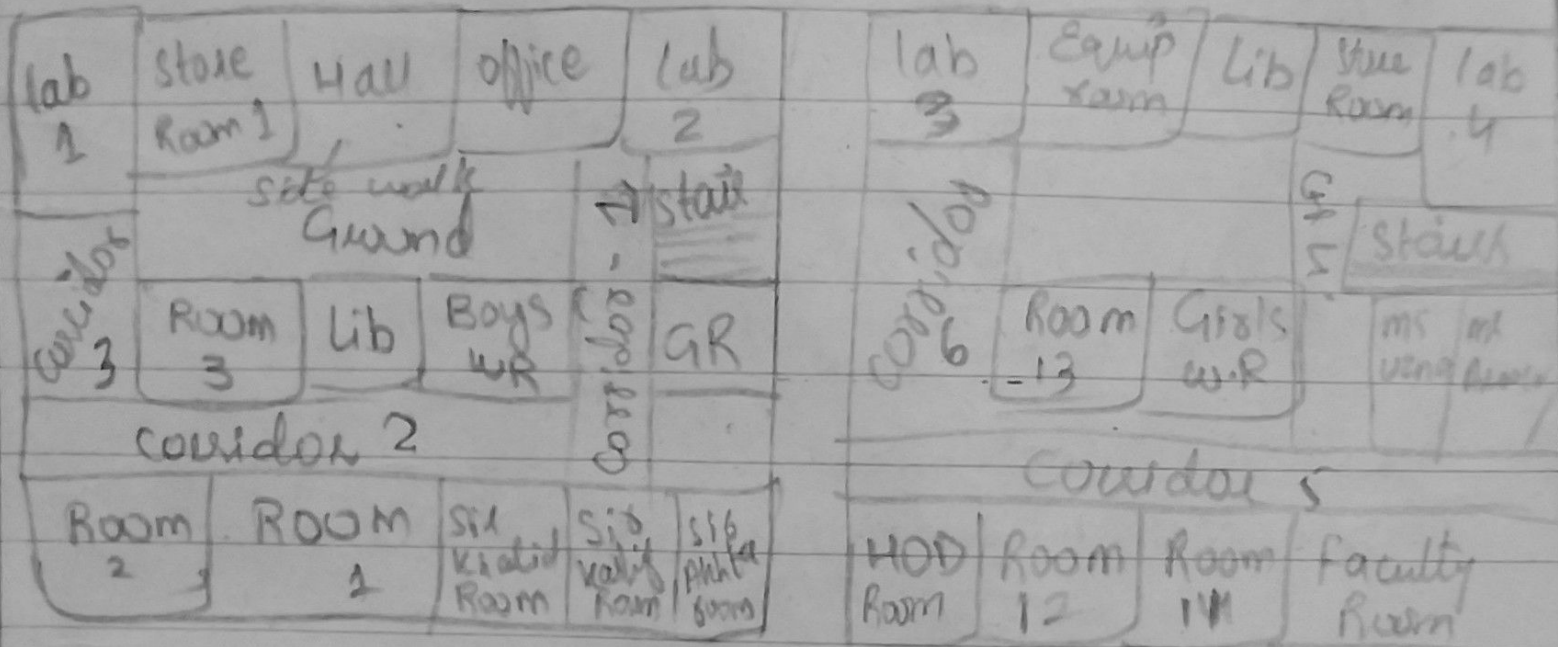


Problem 02.

Ground Floor.

First floor



Problem State:

Initial Point: HOD Room

Goal: Room-2

Graph Representation-

Node

HOD Room

Corridor 05

Corridor 04

Connects To:

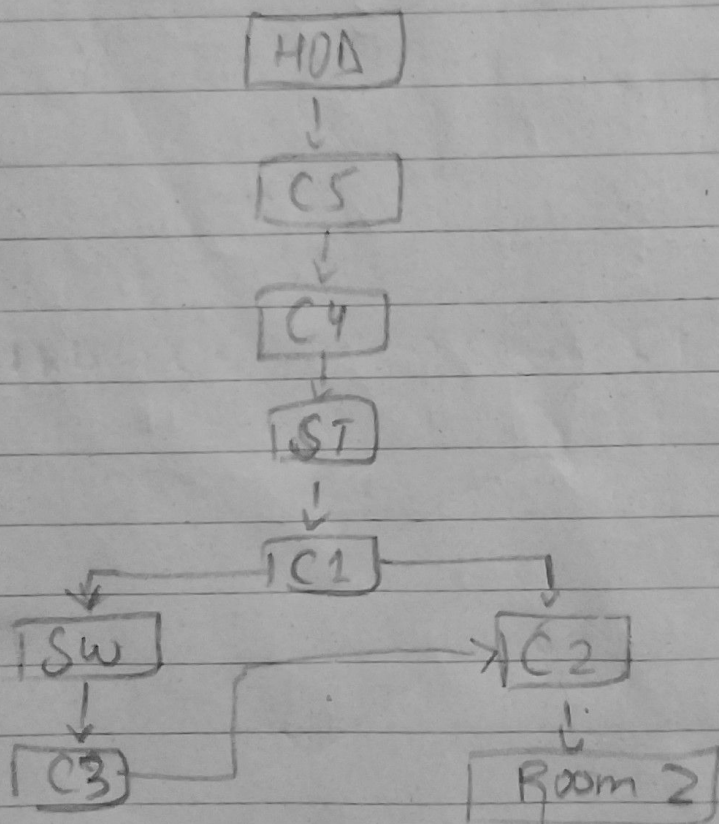
C5

HOD, C4

C5, stair

Stairs	C4, c1
Corridor 1	Stair, c2, Side walk
Corridor - 2	c1, Room 2, c3
Side walk	c1, c3
Corridor 3	Side walk, c2
Room 2	c2

BFS Tree :-



BFS Trace :-

Step 0 :-

$$Q = \{HOD\}$$

$$V = \{HOD\}$$

Step 1: Dequeue HOD Room $\rightarrow$ expand $\rightarrow$ HOD Room $\rightarrow$ Corridor 5.

$$Q = \{C5\}, \quad V = \{HOD, \text{Corridor 5}\}.$$

Step 2: Dequeue Corridor 5 $\rightarrow$ expand Corridor 5 $\rightarrow$ (HOD, Corridor 4)

$$Q = \{C4\}, \quad V = \{HOD, C5, C4\}.$$

Step 3: Dequeue Corridor 4 expand $C4 \rightarrow$ (C5, Stair).

$$Q = \{\text{Stair}\}, \quad V = \{HOD, C5, C4, \text{Stair}\}$$

Step 4: Dequeue Stair $\rightarrow$ expand Stair $\rightarrow$ (Corridor 4, Corridor 1).

$$Q = \{C1\}, \quad V = \{HOD, C5, C4, \text{Stair}, C1\}$$

Step 5: Dequeue C1 expand $C1 \rightarrow$ (C2, Sidewalk, Stair).

$$Q = \{C2, \text{Sidewalk}\}.$$

$$V = \{HOD, C5, C4, \text{Stair}, C1, C2, \text{Sidewalk}\}$$

Step 6: Dequeue C2 expand $C2 \rightarrow$ (C1, Room 2, C3)

$$Q = \{\text{Side walk}, \text{Room 2}, C3\}$$

$$V = \{HOD, C1, C2, \text{Sidewalk}, \text{Room 2}, C3\}.$$

Step 07 : Dequeue Sidewalk, expand Sidewalk

→ (C1, C3) :

Q = { Room 2, C3 }

V = { HOD, ..., Room 2, C3 }

Step 08 : Dequeue Room 2, expand Room 2 → C2 :

Step 9 : Dequeue C3, expand C3 → (C2, Sidewalk) :

All are visited :

Goal Achieved!

Shortest Path :

HOD → C5 → C4 → Stair → C1 → C2 → Room 2

= 6 edges/steps :